1 Introduction

Between 2011 and 2019, Peru's banking agent network—retail establishments authorized to provide basic financial services—expanded from 25% to 85% of all districts, reaching rural areas with no prior financial infrastructure. Yet no quasi-experimental evidence exists on how this expansion affected household finances (AFI 2012). Existing evaluations of agent banking rely on primary data collection (Arraíz 2020, CGAP 2016) or study different contexts such as correspondent banking in Brazil (Fonseca and Matray 2024).

We exploit the staggered rollout across Peruvian districts to estimate causal effects on private remittance flows—a natural first-order outcome in areas where agents provide the only means to receive transfers. Banking agent introduction increases the probability of receiving private remittances by 1.5 percentage points and amounts by 7.2%, with effects concentrated among individuals without bank accounts. This result is robust across heterogeneity-robust DiD estimators (Callaway and Sant'Anna 2021, de Chaisemartin and D'Haultfœuille 2020) but invisible to conventional TWFE—a clear case of the staggered-adoption bias documented by Goodman-Bacon (2021).

2 Data and setting

We combine quarterly administrative data on banking agent presence at the district level from Peru's banking regulator (SBS) with individual-level microdata from the nationally representative household survey (ENAHO), covering 2015Q1–2019Q4. Treatment is defined as the first quarter a district has at least one banking agent.

The estimation sample includes adults aged 18+ in districts that receive their first banking agent during the analysis period, yielding approximately 49,500 individual-quarter observations across 566 districts. Treatment is staggered across 9 cohorts over 20 quarters. Districts that always had agents before 2015 are excluded; districts that never receive an agent during the sample period (91 districts) serve as part of the not-yet-treated comparison group.[1] The sample is predominantly rural (91%), and only one district in the sample received a bank branch during the entire analysis period, confirming that agents represent the sole expansion of formal financial infrastructure in these areas. Baseline financial inclusion is very low (24.7% hold any account). We estimate an intent-to-treat effect, as we observe agent availability but not individual usage. Because SBS data report agent presence quarterly rather than monthly, some measurement error in treatment timing is possible; this would attenuate our estimates toward zero.

3 Empirical strategy

We implement the Callaway and Sant'Anna (2021) difference-in-differences estimator, which computes group-time average treatment effects $ATT(g,t)$ for each treatment cohort g and period t, then aggregates using cohort-size weights. This avoids the "forbidden comparisons"

[1] The CS estimator uses not-yet-treated observations as the primary control group following Callaway and Sant'Anna (2021). The sample difference between CS (N = 40,576) and TWFE (N = 49,506) reflects the balanced cohort requirement of CS.

that bias TWFE estimates under treatment effect heterogeneity—including the possibility of negative weights on some cohorts' effects (Goodman-Bacon 2021). We use the doubly robust inverse probability weighting method of Sant'Anna and Zhao (2020), which provides partial insurance against misspecification of either the outcome or treatment model, controlling for gender, age, rural status, employment, and financial account ownership.

As robustness, we estimate the de Chaisemartin and D'Haultfœuille (2020) dynamic estimator (`did_multiplegt_dyn`), which uses a different weighting scheme and provides joint tests of placebo (pre-trend) coefficients. Because dCDH is estimated without covariates while CS uses doubly robust IPW, differences between the two reflect both weighting and specification choices. We also report conventional TWFE for comparison.

Two validity checks support our design. First, a regression of treatment timing on pre-treatment district-level covariates yields an R^2 of 0.043 (joint F-test $p = 0.103$), indicating that observables explain very little of *when* agents arrive. While some individual covariates differ in levels between early and late cohorts, the doubly robust IPW estimator conditions on these observables, addressing residual imbalance. Second, using demographic characteristics (gender, age, account ownership) as placebo outcomes in both TWFE and CS estimators yields uniformly insignificant effects ($p > 0.15$ in all cases), confirming that sample composition does not change with treatment.[2]

4 Results

4.1 4.1. Main results

Table I presents results across three estimators and samples. Panel A shows effects on remittance receipt probability. The Callaway and Sant'Anna (2021) estimator yields an ATT of 0.015 ($p = 0.021$) in the full sample—with a group-weighted average significant at $p = 0.009$—rising to 0.019 ($p = 0.003$) among individuals without financial accounts. The de Chaisemartin and D'Haultfœuille (2020) estimator confirms the direction, with a first-period effect of 0.022 in the full sample (not individually significant) and 0.034 ($p < 0.05$) among the unbanked. TWFE produces estimates indistinguishable from zero in all specifications, consistent with the attenuation bias predicted by Goodman-Bacon (2021) under staggered adoption with 9 treatment cohorts.

Panel B reports effects on remittance amounts (inverse hyperbolic sine). The Callaway and Sant'Anna (2021) estimates are significant at the 5% level or better across all subsamples, while de Chaisemartin and D'Haultfœuille (2020) estimates point in the same direction but are imprecisely estimated.

[2]TWFE placebo p-values: gender 0.91, age 0.45, rural 0.24, account 0.42. CS placebo p-values: gender 0.56, age 0.16, account 0.81.

Table I. Effect of Banking Agent Introduction on Private Remittances

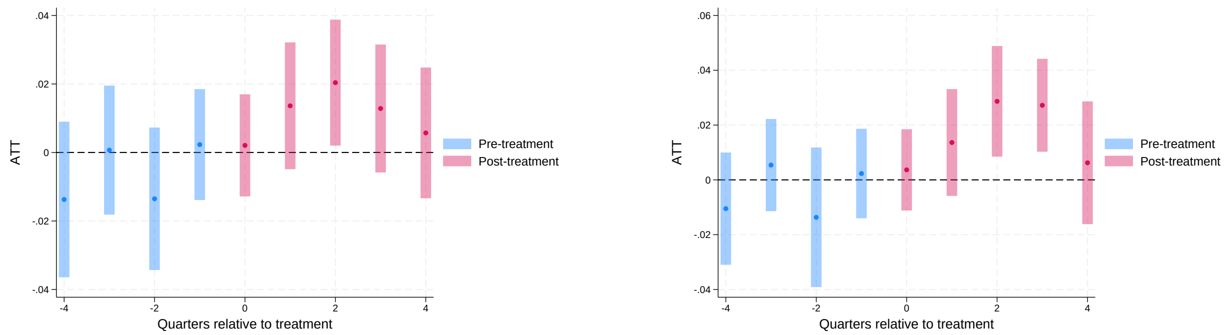
	Full Sample			Without Account		
	CS	dCDH	TWFE	CS	dCDH	TWFE
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Receives remittances (indicator)						
ATT	0.0152**	0.0223	−0.0002	0.0186***	0.0340**	−0.0010
	(0.0066)	(0.0131)	(0.0037)	(0.0064)	(0.0166)	(0.0040)
Panel B: Remittance amount (asinh)						
ATT	0.0718**	0.0603	0.0016	0.0793***	0.1148	−0.0029
	(0.0302)	(0.0519)	(0.0183)	(0.0287)	(0.0658)	(0.0195)
Controls	Yes	No	Yes	Yes	No	Yes
N	40,576	49,526	49,506	30,688	37,292	37,292

Notes: CS = Callaway and Sant'Anna (2021) aggregated ATT using doubly robust IPW (Sant'Anna and Zhao 2020). dCDH = de Chaisemartin and D'Haultfœuille (2020) first-period dynamic effect, estimated without covariates. TWFE = two-way fixed effects with district and quarter FE, clustered at district level. CS and TWFE controls: gender, age, rural, employment, account ownership, account×employment. Survey weights applied to CS and TWFE. Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.2 4.2. Pre-trends and event study

Figure 1 presents event study estimates for remittance receipt. Pre-treatment coefficients are small and insignificant in both samples, supporting parallel trends. Post-treatment effects emerge around the third quarter. Additionally, the dCDH joint placebo test yields $p =$ 0.37–0.53 across specifications,[3] and a permutation-based placebo—randomizing treatment dates 100 times—produces significant estimates in fewer than 15% of iterations. Among account holders, neither CS nor dCDH yields significant effects (not reported), consistent with existing account holders having alternative channels for remittance receipt.

[3] Joint placebo p-values: receives remittances 0.371 (full) / 0.460 (no account); remittance amount 0.356 (full) / 0.534 (no account).



(a) Full sample (b) Without financial account

Figure 1. Event study: probability of receiving remittances

Notes: Callaway and Sant'Anna (2021) event study estimates with 95% confidence intervals. Reference period: $e = -1$. Doubly robust IPW (Sant'Anna and Zhao 2020). Survey weights applied.

5 Discussion

Why does TWFE yield a zero estimate while CS and dCDH do not? The answer lies in treatment effect heterogeneity across cohorts. Cohort-specific ATTs range from -0.056 to $+0.074$, with three cohorts individually significant at the 5% level—all positive.[4] TWFE assigns variance-based weights—potentially negative—to these heterogeneous effects, which can produce sign-reversed or attenuated estimates rather than a meaningful average (Goodman-Bacon 2021). CS and dCDH restrict comparisons to treated-versus-not-yet-treated with non-negative weights, recovering the correctly signed aggregate effect. The CS group-weighted average is significant at $p = 0.009$. Results are robust to excluding the largest expansion cohort (2017Q2, 64 districts): the ATT remains significant at $p = 0.05$ for remittance receipt and $p = 0.04$ for amounts.[5]

What do agents actually do for rural households? Our results suggest they serve as *cash access points* rather than gateways to formal banking—we find no effects on account ownership, remittance sending, cash usage, or labor income (available upon request). Whether remittances flow directly through the agent or the agent's presence simply facilitates transfers via other channels remains an open question with these data.

The effect magnitude deserves comment. A 1.5–1.9 pp increase on a 4.6% base implies a 33–41% increase in remittance receipt—large for an ITT, but not implausible in districts that had *zero* financial infrastructure before the agent arrived. Burgess and Pande (2005) and Jack and Suri (2014) document comparable magnitudes for bank branches in India and mobile money in Kenya, albeit for different outcomes and under stronger identification designs.

[4]Significant cohorts span the analysis period: 2015Q3 (56 districts, 5,245 obs), 2016Q4 (5 districts, 452 obs), and 2018Q1 (18 districts, 1,216 obs). Under the null of no effect, one would expect $\sim$0.85 significant cohorts out of 17 at the 5% level; observing three, all positive, is unlikely by chance. One cohort (2016Q2, 33 districts) shows a marginally negative effect but does not differ in observable characteristics.

[5]Excluding 2017Q2: ATT for remittance receipt $= 0.013$ ($p = 0.050$); for amounts $= 0.064$ ($p = 0.043$).

6 Conclusion

Banking agents in rural Peru increase private remittance receipt, particularly for the unbanked. The effect is robust to estimator choice within the class of heterogeneity-robust DiD methods but invisible to TWFE. These findings speak to the specific context of previously unbanked rural districts and should not be extrapolated to settings with pre-existing financial infrastructure.

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